

Subtitle Basic idea about DC to AC converter
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Subject Power Electronics EE 603

Inverter Lecture 03

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Hello friends

Lecture 03

- ① Brief recap of last lecture 02
- ② A brief review of Fourier series
- ③ inverter performance parameters
- ④ Numerical example
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A brief review of Fourier series

- Fourier series is an infinite series for representation of a periodic function in the terms of trigonometric sine and cosine functions.
- A function $f(x)$ is said to be a periodic if $f(x + T) = f(x)$ for all real x and some positive number T . T is known as the period of $f(x)$
- Trigonometric series:

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Where a_0 , a_n and b_n ($n = 1, 2, 3, \dots$) are called co-efficient.

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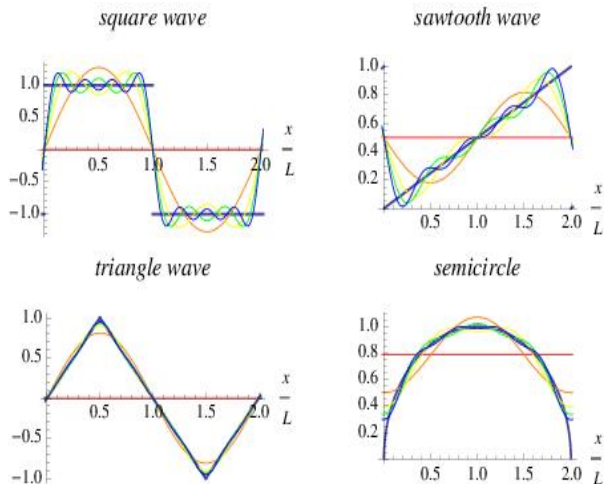


Figure: wave 1

NB:<https://mathworld.wolfram.com/FourierSeries.html>

cont...

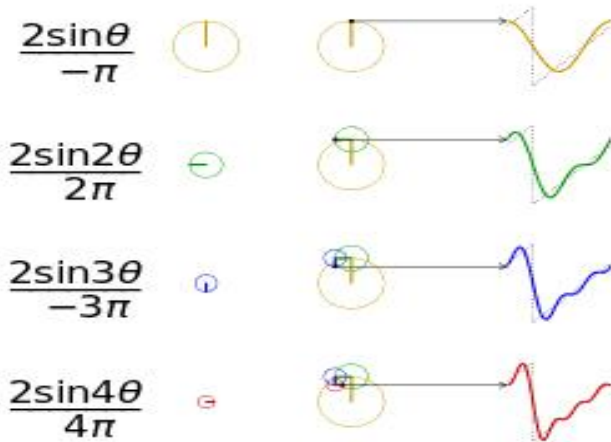


Figure: wave 2

NB:<https://electronics.stackexchange.com/questions/148288/>

Inverter performance parameters

- Harmonic factor of n^{th} harmonic (HF_n)

$$HF_n = \frac{V_{on}}{V_{o1}}; n > 1 \quad (1)$$

- Total harmonics distortion (THD)

$$THD = \frac{1}{V_{o1}} \left(\sum_{n=2,3,4,5\dots}^{\infty} V_{on}^2 \right)^{1/2} \quad (2)$$

- Distortion factor (DF)

$$DF = \frac{1}{V_{o1}} \left(\sum_{n=2,3,4,5\dots}^{\infty} \frac{V_{on}^2}{n^2} \right)^{1/2} \quad (3)$$

cont...

- Lowest order harmonics (LOH) 3% of the fundamental

Numerical example

• A 1-ph half-bridge inverter has a resistive load of $R = 2.4\Omega$ and DC input voltage $V_s = 48V$; Find out (a) rms output voltage of the fundamental V_{o1} (b) the output power P_o (c) the ave and peak current of each transistor (d) the peak reverse blocking voltage V_{BR} of each transistor (e) the THD (f) The DF

• Solv.

$$(a) V_{o1} = 0.45 * V_s = 0.45 * 48 = 21.6V$$

$$(b) V_o = \frac{V_s}{2} = \frac{48}{2} = 24V$$

$$P_o = \frac{V_o^2}{R} = 240W$$

$$(c) I_{peak} = \frac{V_o}{R} = 10A \text{ therefore, } I_{Qave} = duty * I_p = 5A \text{ as duty} = 50\%$$

$$(d) V_{BR} = 2 * V_o = 48V$$

cont...

(e) $V_n = 0.2176V_s$ and $V_{o1} = 0.45V_s$ therefore,

$$\text{THD} = \frac{0.2176V_s}{0.45V_s} = 48.34\%$$

$$(f) \text{DF} = 5.382\%$$

References



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